NOT MEASUREMENT SENSITIVE

MIL-T-62355B
6 November 1991
SUPERSEDING
MIL-T-62355A
19 November 1985

MILITARY SPECIFICATION

* TRUCK, MAINTENANCE: TRUCK, VAN: AND TRUCK, FLATBED:
WITH TELESCOPIC OR ARTICULATING BOOM ONE-MAN AERIAL BASKET,
7600 TO 9140 MILLIMETER (25 TO 30 FOOT),
MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers gasoline or diesel engine driven telescopic or articulating boom one-man aerial basket trucks. Trucks are of three body styles: maintenance, forward control van, and flatbed.

* 1.2 Classification. The vehicle shall be one of the combinations of type, class, body style, and size shown in table I, as specified (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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* TABLE I. Vehicle classification.

Type	Boom description	Class	Body style	Size			
I	Telescopic boom.	A	2	25	27	--	--
II	Articulating boom with limited angle travel (minimum 0-125 degrees travel between booms).	B	1 or 3	--	27	28	--
		C	1 or 3	--	--	28	30
III	Articulating boom with limited angle travel (minimum 90-155 degrees travel between booms).	A	2	--	27	--	--
		B	1 or 3	--	27	--	--
		C	1 or 3	--	--	--	30
IV	Articulating boom with overcenter angle travel (minimum 0-180 degrees between booms).	B	1 or 3	--	27	--	--

* 1.2.1 Class. The vehicle shall be one of the following classes:

Class A - 3858 kg (8,500 pounds) minimum GVW (style 2)
Class B - 4630 kg (10,200 pounds) minimum GVW (styles 1 or 3)
Class C - 8350 kg (14,000 pounds) minimum GVW (styles 1 or 3).

1.2.2 Style. The vehicle shall be one of the following body styles:

Style 1 - Maintenance body
Style 2 - Van body with interior shelves
Style 3 - Flatbed.

1.2.3 Size. The size of the vehicle shall be determined by the maximum height the aerial basket can attain, measured from ground level to the basket floor. Sizes shall be the minimum sizes specified in table II.

TABLE II. Size.

Size	Ground to basket floor height	
	mm	feet
25	7600	(25)
27	8230	(27)
* 28	8534	(28)
30	9140	(30)

1.2.4 Model. The model of the vehicle shall be determined by the number of wheels and the number of driving wheels, as specified (see 6.2).

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Model 4x2 - 4 wheels, 2 wheels driving (style 1, 2 or 3)
Model 4x4 - 4 wheels, 4 wheels driving (style 1 or 3 only).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

*	MIL-P-514	- Plates, Identification, Instruction, and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for.
	MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
	MIL-H-83282	- Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO code Number H-537.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

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MS 51317 - Light, Warning, Vehicular, Rotating, D.C.
MS 51335 - Pintle Assembly, Towing - 18,000 Lbs. Capacity, Manual Release.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

* (Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

* (Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.
1926.556 - Aerial Lifts

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* (Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI A92.2	- American National Standard for Vehicle Mounted Elevating and Rotating Aerial Devices.

(Applications for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

	ASTM B 117	- Standard Method of Salt Spray (Fog) Testing (DoD adopted).
*	ASTM D 4985	- Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

* (Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices.

J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).

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	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
*	J683	- Tire Chain Clearance - Trucks, Buses, and Combination Vehicles.
	J704	- Openings for Six-and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DOD adopted).
	J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Applications for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless an specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.13, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.13, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data.

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Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. The exterior color shall be gloss yellow matching color chip no. 13538 of FED-STD-595. Extruded, molded, or spiral wound booms and baskets need not be painted.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Warning plates. Warning plates shall be furnished and mounted in a conspicuous location near the lower and upper controls and in the driver's compartment. Plates shall conform to composition A (Class 1 or 2) or composition C of MIL-P-514. At least the following information shall appear on each of the three plates:

WARNING

(a) AERIAL BASKET MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED BASKET LOAD, NOT IN EXCESS OF _____ POUNDS.
(c) INSPECT BASKET LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE BOOM AND BASKET WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WITH BOOM(S) IN STOWED POSITION _____.
(f) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 48 kilometers/hr (km/hr) (30 MPH).
(g) BASKET LINER MUST BE INSTALLED WHEN OPERATING ON OR NEAR ELECTRICAL LINES.
(h) DO NOT PERMIT DIRECT CONTACT OF BASKET TO GROUND WHEN OPERATING ON OR NEAR ELECTRICAL DISTRIBUTION LINES.
(i) MAXIMUM LONGITUDINAL INCLINE _____ DEGREES.
(j) MAXIMUM TRANSVERSE SLOPE _____ DEGREES.
(k) BOOM(S) AND BASKET MUST BE KEPT CLEAN AND FREE OF DAMPNESS.
(l) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER.
(m) USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID.
* (n) WIRE BRAID ELECTRICAL HOSES MUST NOT BE USED ON OR NEAR ELECTRICAL LINES.

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* 3.1.1.4 Instruction plates. An instruction plate showing a schematic of all valves, valve controls, and aerial equipment, and complete operating instructions for operating the aerial equipment shall be fastened on the inside of a curbside vertical cabinet door for maintenance trucks and on the inside of the rear door for vans. Each control lever shall be marked with an instruction plate depicting the function of the control. Individual markings shall be adjacent to each control. Plates shall conform to composition A (Class 1 or 2) or composition C of MIL-P-514.

3.1.1.5 Warning signs. Warning signs or labels, as recommended by the Manufacturers of Aerial Devices and Digger Derrick Council, shall be furnished and attached.

3.1.1.6 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.7 Drain plugs. All drain plugs installed in manual transmissions, transfer cases and rear axles shall be of the permanent magnet type.

* 3.1.1.8 Silicone brake fluid. When specified (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A decal or plate shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176." The decal or plate shall conform to Type I or Type II, Class I of MIL-M-43719 or composition A (Class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.9 Trailer towing package. When specified (see 6.2), a trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51335-1. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 815 kg (1,800 pounds) and a horizontal drawbar load of not less than 45 kilonewtons (kN) (10,000 pounds) directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than 100 mm (4 inches) forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment device shall be capable of accommodating a standard grab hook [56 mm (2-3/16 inches) wide, 18.7 mm (47/64 inch) thick and 12.3 mm (31/64 inch) throat width)] for a 9.5 mm (3/8-inch) chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, or number coded, shall be mounted in a readily accessible location near the pintle.

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3.1.1.10 Drive hubs. When specified for model 4x4 vehicle (see 6.2), manually operated drive hubs for disengaging the front wheels from the drive train shall be provided, except for vehicles equipped with interaxle compensating devices between the front and rear axles (see 3.4.5.1.1).

3.1.1.11 Brushguard. Model 4x4 vehicles shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed in and protected by the front bumper, a headlamp brushguard is not required.

3.1.1.12 Rear wheel splash and stone throw protection. Rear wheels shall be provided with splash and stone throw protection in accordance with SAE J682.

* 3.1.1.13 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Carrier Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with the State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94.

* 3.2.4 Safety standards. The vehicle, aerial device manufacturer, other contributing manufacturer's, dealers (when applicable) and installers (when applicable) shall comply with ANSI A92.2. The vehicle shall comply with all Occupational Safety & Health Administration requirements, including OSHA 1926.556.

* 3.2.5 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body and aerial platform assembly in stowed position; full complement of fuel,

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lubricants and coolant. As evidence of conformance to 3.6.4.1, the curb weight of the completed vehicle shall be not greater than that specified in table III.

TABLE III. Maximum curb weight.

	Class	Style	Maximum Curb weight Kg (Pounds)
	A	2	3630 (8,000)
*	B	1	4445 (9,800)
	B	3	4080 (9,000)
*	C	1	5450 (12,000)
*	C	3	5220 (11,500)

3.2.6 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 80 kg (175 pounds)), and a payload distributed evenly throughout the length of the vehicle body, to provide not less than the required GVW.

3.2.7 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

* 3.2.8 Static loads. The maximum static load on any axle, suspension system component or tire shall not exceed twice the dynamic rating of that component, with the vehicle and aerial basket fully loaded and placed in any position the aerial boom(s) are capable of attaining.

3.2.9 Rating. Vehicle and component ratings shall be the manufacturer's published ratings. Vehicle and component ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of the ratings must be submitted to the engineering office of the procuring activity. Minimum GVW ratings shall be not less than specified for the specified class of the vehicle. The ratings shall be those established by the manufacturer of the equipment being verified.

* 3.2.10 Dimensions. The cab-to-axle dimension shall be not less than 1520 mm (60 inches) for class B and class C, styles 1 and 3 maintenance and flatbed trucks. The cab-to-axle dimension shall be measured from the back face of the cab to the center of the axle. The wheelbase for class A, style 2 vans shall be not less than 3175 mm (125 inches). The tread dimensions for styles 1 and 3 maintenance and flatbed trucks shall be not less than 1620 mm (64 inches) at the front and not less than 1575 mm (62 inches) at the rear. The tread dimensions for class A, style 2 vans shall be not less than 1730 mm (68 inches) at the front and not less than 1676 mm (66 inches) at the rear. The tread dimensions shall be measured between the tire centerlines, or between the centerlines of the tire and wheel assemblies when dual rear wheels are furnished.

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accept the defrosters, and which installed no water shall leak into the vehicle

3.2.11 Accessibility. The design of the vehicle and optional equipment shall allow access for routine servicing, and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.12 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance.

3.3.1 Speeds and gradeability. The vehicle shall maintain a high speed of not less than 105 km/h (65 miles per hour (mph)) on smooth, approximately level, hard surfaced roads. The vehicle shall be capable of ascending grades of at least 20 percent in low speed range. Speed and gradeability requirements shall be met with the vehicle loaded to required GVW.

* 3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.3.3 Parking brakes. A hand- or foot-actuated parking brake shall be furnished in addition to the service brakes. When a foot actuated parking brake is furnished, a hand lever shall be furnished to release the parking brake. The vehicle parking brake shall be capable of holding the vehicle loaded to specified GVW on a 30 percent slope.

* 3.3.4 Upsetting stability. With one and one half times the specified load (see 3.6.5) placed in or suspended from the aerial basket, with the vehicle otherwise unloaded, the boom(s) shall be capable of placing and holding the loaded aerial basket in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one half times the specified load when any one of the vehicle tires lifts off the ground, ANSI A92.2 not withstanding. The stability test shall not cause any permanent deformation beyond the elastic limit of any component of the vehicle.

* 3.3.4.1 Stability on slopes. With one and one third times the specified load (see 3.6.5) placed in or suspended from the basket, with the vehicle otherwise unloaded, with the vehicle on a 5-degree slope, with the outriggers (if furnished) extended, without the vehicle being leveled (ANSI A92.2 not withstanding), facing upward, then downward, then at right angles to the 5-degree slope, the boom(s) shall be capable of placing and holding the loaded aerial basket in any attainable position of the equipment without

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part of the vehicle. [illegible] The vehicle

causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one third times the specified load when any one of the vehicle's tires or outriggers lifts off the ground, ANSI A92.2 not withstanding.

3.3.5 Wind resistance. With the specified basket load (see 3.6.5) and the vehicle otherwise unloaded and in any extended position, the vehicle shall be capable of withstanding a wind load of 335 Newtons per square meter (N/m^2) (7 pounds per square foot) without upsetting, overturning, or permanently deforming any component. The maximum area exposure for wind load shall be calculated at the body, boom(s), and basket in the most critical position that would induce tipping. The resisting moment shall be greater than the wind load when the load is applied on the surface of the body, boom, and basket. The contractor shall perform calculations or have available a certified test report for the vehicle or identical group of the same type, class, style and size of vehicle.

* 3.3.6 Height. The boom(s) shall be capable of raising the aerial basket, with rated load, from ground level to the minimum height specified in table II for the size of the vehicle furnished. Height shall be measured from ground level to basket floor.

* 3.3.7 Horizontal reach. The boom assembly shall provide the minimum horizontal reach specified in table IV for the type, size, and class of vehicle, measured from the front edge of the aerial basket to the vertical axis of the turntable. Horizontal reach shall be measured with the boom fully extended and in a horizontal position for type I, and with the booms positioned so that the aerial basket is at its maximum attainable side extension for types II, III, and IV. The boom(s) shall be capable of lowering the loaded aerial basket to within 915 mm (three feet) of the ground for type I, and to within 1220 mm (four feet) of the ground for types II, III, and IV. The time needed to move the boom assembly from stowed position to maximum horizontal reach position shall be within the time required by table IV.

TABLE IV. Horizontal reach.

Type	Class	Size	Horizontal reach		Time required (seconds)	
			mm	feet	(min.)	(max.)
I	A	25	5486	(18)	15	70
	A	27	5944	(19.5)	15	70
II	B	27, 28	4880	(16)	15	70
	C	28, 30	5486	(18)	15	70
III	A, B	27	4880	(16)	15	70
	C	30	5486	(18)	15	70
IV	B	27	7010	(23)	15	70

3.3.8 Rotation (aerial basket). The boom assembly shall be capable of rotation in either direction in the horizontal plane. Rotation may be either continuous or non-continuous; except, when specified (see 6.2), continuous

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rotation shall be furnished. The turntable motor shall rotate the loaded basket through an arc of 360 degrees in not less than 30 seconds and not more than 75 seconds. The turntable rotating device shall remain in a fixed position when the hydraulic motor is de-energized. The turntable shall be capable of smoothly rotating the aerial assembly up and down a 5 percent grade when loaded to rated working capacity, and producing the maximum attainable tipping moment.

3.3.9 Position stability (aerial basket). When the vehicle is on a 5 percent grade, without leveling the vehicle, the elevating and rotating mechanism shall hold the loaded aerial basket (see 3.6.5) in any extended position within 25 mm/hr (1 inch per hour) horizontal creep and 50 mm/hr (2 inches per hour) vertical settling. The creep and settling shall be measured at the front of the aerial basket with hydraulic fluid stabilized at ambient air temperature.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Gasoline engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be furnished with a liquid cooled, internal combustion, four-stroke cycle, gasoline engine. The engine shall develop not less than 8 kw (115 net brake horsepower) at manufacturer's recommended governed speed and shall have a net torque capacity of not less than 290 Newton meter (215 pound feet). Engine net torque and net power rating shall be determined in accordance with SAE J245 or SAE J1349. The engine shall develop the specified torque and power when operated on unleaded fuel with a research octane rating of 91. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual.

* 3.4.1.2 Diesel engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net power shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

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* 3.4.1.3 Coolant level indicator. A coolant level indicator consisting of an unbreakable translucent reservoir shall be provided in a position readily visible for checking coolant level. The translucent reservoir shall be part of a coolant recovery system.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 3050 m (10,000 feet) above sea level or in an ambient air temperature of not less than 52°C (125 degrees Fahrenheit (°F)).

* 3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.7 Power plant heaters. When specified (see 6.2), a coolant heater, an engine oil heater and fuel warmers (diesel engine driven vehicles only) shall be provided. Heaters shall operate on 120 volts alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as practicable. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to -51°C (-60°F)) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation. Heaters and fuel warmers shall be furnished as follows:

(a) A coolant heater, 1000-watt (W) minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 77°C (170°F) to 90°C (195°F) shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

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* (1) Not less than 16 W/L (15 watts per quart) heating capacity
* (2) Not more than 2.8 watts per square centimeter (W/cm^2) (18 watts per square inch) heating capacity
(3) Thermal balance design or thermostat control providing for uninterrupted operation
(4) Provision for mounting below minimum service oil level.

* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27°C (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.7(c) and (d) may be combined.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Ignition system. For gasoline engine driven vehicles, a 12-volt direct current (dc) ignition system shall be furnished.

* 3.4.2.2 Starting system. For diesel engine driven vehicles, a 12- or 24-volt dc starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or electric glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (ml) (12 fluid ounces) shall be furnished. A reservoir need not be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), a minimum 85-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 40 amperes.

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* 3.4.2.4 Battery(s). Each battery shall be of 12-volt potential. The reserve capacity ratings and the total cold cranking ratings at -18°C (0°F), both measured in accordance with SAE J537, shall be not less than specified in table V. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE V. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	135	550
Gasoline	80	350

* 3.4.2.5 Lighting. All vehicle lights and reflectors shall conform to Federal Motor Carrier Safety Regulations 393.11, 393.12 (when vehicle exceeds 2030 mm (80 inches) in width), 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.5.1 Turn signals. Turn signals shall be furnished. Turn signal lights may be combined with stoplights at rear and with parking lights at front. Taillights and stoplights may be incorporated into the same housing. Control for turn signals shall be self-cancelling type, with visible flash indicators. Turn signal lamps shall conform to SAE J588. Turn signal operating unit shall conform to SAE J589, class A, and shall be mounted on the steering column.

* 3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67 (except compliance with 393.67(c), subparagraphs (1), (4)(ii), (11), 393.67(d) and 393.67(f) is not required). The capacity of the fuel tank shall be not less than 72 L (19 gallons).

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust shall discharge to the atmosphere to the rear of the vehicle or to the side behind the rear wheels. A spark arresting muffler shall be furnished on model 4x4 vehicles and shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

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3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission on style 1 maintenance and style 3 flatbed trucks shall have not less than four forward speeds and one reverse speed. The transmission on style 2 vans shall have not less than three forward speeds and one reverse speed. The transmissions shall provide for maximum ease of shifting with synchronized shift for at least the two highest forward speeds. Manual transmissions on style 1 maintenance and style 3 flatbed trucks shall be provided with a power takeoff openings conforming to SAE J704 (see 3.4.5.4).

* 3.4.5.1.1 Transfer case (4x4 model). A two-speed transfer case shall be provided on model 4x4. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of two-wheel or four-wheel drive. When interaxle compensating devices are furnished, they shall provide for positive transfer of power to all the driving axles. The speedometer shall read accurate speed km/h (mph) with the speed selector in high or low range.

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle class and model and for the size engine furnished, with torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free.

* 3.4.5.3 Automatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic transmission. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than three forward gear ratios. Normal driving range selector position shall provide not less than three gear ratios without movement of the selector. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

* 3.4.5.4 Power takeoff (styles 1 and 3 maintenance and flatbed trucks only). Style 1 maintenance trucks and style 3 flatbed trucks, when furnished with a manual transmission, shall have a power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type, style, class and model of vehicle furnished. Style 2 vans may be of integral body-frame construction. When additional torsional rigidity is required, the vehicle shall have frame reinforcement. Reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment and torsional resistance to at least equal the torsional loads imposed with the aerial basket fully loaded and placed in any attainable position and with the vehicle loaded to the specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a total rated capacity per axle at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at each wheel. Suspension stabilizer devices shall be furnished. Suspension system ratings shall be adequate for the loads imposed during all operating conditions when in a static position.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. A single reduction rear axle shall be furnished.

3.4.9.1 Traction control. When specified (see 6.2), traction control shall be furnished on the rear axle. Traction control shall be of the positive traction, limited slip or automatic locking differential type.

* 3.4.10 Wheels, rims, tires, and tubes. Class A vehicles shall be equipped with single front and single rear wheels. Class B and C vehicles shall be equipped with single front and dual rear wheels. Rim sizes and disc wheel sizes shall be the same for all wheels on the vehicle. Tire size and load range (ply rating) shall be the same for all tires on the vehicle. Rim and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished.

* 3.4.10.1 Tires. Tires shall be of the tubeless truck type with highway tread for 4x2 model vehicles and with non-directional mud and snow tread for 4x4 model vehicles. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of the steel belted radial type or when specified (see 6.2), bias ply. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW (see 3.2.5). Tires shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organization recommendations.

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* 3.4.10.2 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.4.15. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.4.10.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.4), a spare wheel or rim shall be furnished.

* 3.4.10.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

3.4.11 Brakes. Manufacturer's standard power brakes shall be furnished. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42, 393.45 through 393.48, and 393.52. Brake linings shall be of nonasbestos material.

3.4.12 Steering. Power steering shall be furnished.

* 3.4.13 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and washers. Windshield wipers shall be multispeed type and operated by electric motor(s). When specified (see 6.2), intermittent wipers shall be furnished.

3.4.14 Bumpers. Front and rear bumpers shall be provided. Rear end protection shall conform to Federal Motor Carrier Safety Regulation 393.86. For style 1 maintenance trucks and style 3 flatbed trucks, a safety-tread steel plate step, integrated with the rear bumper, extending across the width of the body and with a step surface not less than 250 mm (10 inches) deep, shall be provided.

* 3.4.15 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a jack, jack handle, and wheel-nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with a flat tire. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly. Stowage space for these tools may be furnished in

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the vehicle when stowage space for these tools is located outside the vehicle, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

* 3.4.17 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted and visible to the driver when seated in the driving position.

3.4.18 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Ash receptacle
(h) Dual sunvisors
* (i) Front door mounted armrest on driver's and passenger's side
(j) Driver's compartment ventilator other than window
(k) Domelight switch operable from driver's seat
(l) Door operated domelight switch for both doors
* (m) Tachometer (class B and C trucks only, with manual transmission only).

* 3.4.19 Rearview mirrors. An outside rearview mirror shall be provided on both sides. The outside rearview mirrors shall be of the Junior West Coast or low mount type incorporating a wide angle viewing system. The upper mirror shall have a flat reflective area of not less than 230 square centimeters (35 square inches). The lower mirror shall have a convex reflective area of not less than 160 cm^2 (25 square inches) and a radius of curvature or not less than 510 mm (20 inches). The mirror assembly heads shall be adjustable and mounted to minimize obstruction of forward lines of vision.

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3.4.20 Horn. The manufacturer's standard electric horn shall be furnished.

* 3.4.21 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

* 3.4.22 Servicing and adjusting. Prior to acceptance of each vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete servicing of chassis, engine, running gear and mounted equipment with grades of lubricants and hydraulic fluids recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume, and servicing of the windshield washer reservoir with water and appropriate additives. Cables, where used, shall be lubricated with proper lubricants to resist corrosion.

3.5 Body styles.

3.5.1 Style 1 maintenance trucks. Style 1 maintenance trucks shall be equipped with a cab and a line maintenance and construction type body having the minimum dimensions specified in table VI. The body shall include weatherproof cabinets on each side of a center loadspace. The turntable pedestal of the boom shall be securely mounted to the truck chassis frame either directly or through a suitable subframe, and shall be located in the loadspace. The subframe and floor shall not be welded to the truck chassis frame. The loadspace floor shall be of not less than 14 gage (1.897 mm) (0.0747 inch) safety tread steel plate or equivalent aluminum. Any area of 930 cm^2 (one square foot) shall be capable of sustaining a live load of 226 kg (500 pounds) in the loadspace and 45 kg (100 pounds) in the cabinets and on shelves without permanent deformation.

TABLE VI. Body dimensions (minimum).

Features	Class A		Class B		Class C	
	mm	inches	mm	inches	mm	inches
Overall length of body	2440	(96)	2590	(102)	2590	(102)
Loadspace:						
Width	1168	(46)	1168	(46)	1168	(46)
Height	660	(26)	660	(26)	660	(26)
Cabinet space:						
Length	2440	(96)	2590	(102)	2590	(102)
Height	940	(37)	940	(37)	940	(37)
Width	356	(14)	356	(14)	356	(14)
	m^3	in^3	m^3	in^3	m^3	in^3
Volume (each side)	0.6	(37,000)	0.65	(40,000)	0.65	(40,000)

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* 3.5.1.1 Cab. The manufacturer's standard full width cab shall be furnished. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with both doors equipped with external, key-operated locks. Drip rails shall be provided above the cab doors. Unless otherwise specified, the cab shall have upholstered full-width adjustable seat and back or individual adjustable driver's seat and individual passenger's seat. When specified (see 6.2), the cab shall be equipped with an individual, adjustable driver's seat and individual passenger's seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior dome lighting shall be provided. Seat belts shall be installed. Outboard seats shall have combination pelvic and upper torso restraint seat belts.

* 3.5.1.2 Bulkhead. The bulkhead shall be constructed to a height of not less than 560 mm (22 inches) above the floor level and shall be of not less than 18 gage (1.214 mm) (0.0478 inch) high tensile steel or 16 gage (1.519 mm) (0.0598 inch) carbon steel, reinforced for rigidity or equivalent aluminum. The bulkhead shall be capable of withstanding a horizontal static load equal to one half the payload capacity of the vehicle without permanent deformation.

* 3.5.1.3 Tailgate. The tailgate shall be constructed of not less than 18 gage (1.214 mm) (0.0478 inch) high tensile steel or 16 gage (1.519 mm) (0.0598 inch) carbon steel, reinforced for rigidity or equivalent aluminum. The tailgate shall be hinged and equipped with supports for retention in a horizontal position, flush or slightly lower than the floor level.

* 3.5.1.4 Cabinets. Cabinet headers shall be not less than 18 gage (1.214 mm) (0.0478 inch) high tensile steel or 16 gage (1.519 mm) (0.0598 inch) carbon steel. Cabinet backs, partitions, and shelves shall be of not less than 20 gage (0.919 mm) (0.0359 inch) high tensile steel or 18 gage (1.214 mm) (0.0478 inch) carbon steel. Cabinet doors shall be of not less than 20 gage (0.919 mm) (0.0359 inch) high tensile, or 18 gage (1.214 mm) (0.0478 inch) carbon steel, single thickness reinforced, or 22 gage (0.759 mm) (0.0299 inch) carbon steel, double panel, reinforced. Aluminum cabinets of equivalent strength may be furnished. Doors shall be weatherproof and constructed to maintain fit and alinement in service. Doors shall have continuous rod, or hem type, heavy duty hinges, with noncorrosive hinge pins or sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of, or plated with, corrosion-resistant material. Horizontal doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 1700 kn on any square meter (25 pounds on any square foot) of area. Cabinet arrangement shall be manufacturer's standard and shall provide the following:

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(a) Full height vertical cabinets equipped with bolted or screwed-on hooks
(b) Horizontal cabinets equipped with small parts bins with adjustable spacers, shelves and partitions, and a rubber-goods compartment
(c) Forward cabinets equipped with not less than two adjustable tray type shelves.

* 3.5.1.5 Maintenance body lighting. In addition to the lighting specified in 3.4.2.5, the following lighting and wiring shall be furnished:

(a) Lights with toggle switches and guards in each cabinet compartment;
(b) Lights on universal mounts and of sufficient candlepower to illuminate body loadspace and all equipment contained therein and all controls;
(c) Two 115 mm (4-1/2 inch) minimum diameter sealed beam floodlights with universal mounts, one on each side on the top of the cab, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area;
* (d) An amber rotating vehicular warning light in accordance with MS 51317-4. The rotating light shall be mounted with reinforcement at the point of attachment on or near the left front corner of the cab, body, boom rest or other support for 360 degree visibility. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch;
(e) A demountable projecting load light on the end of the boom, with a service line and a socket when the end of the boom extends beyond the rear of the vehicle. A receptacle on the rear of the body for the light shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18;
(f) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292. All wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body framework.

3.5.1.6 Body mounting. The body shall be mounted in accordance with the manufacturer's recommendations. Breaker strips, when used, shall be tapered toward the front of the vehicle.

3.5.2 Style 2 van body. The style 2 van body shall be a forward control type with integral cab and body. The van shall be of all metal construction having provisions for doors and windows specified herein. The body shall have steel side panels without side windows. Drip rails shall be installed above the doors.

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* 3.5.2.1 Driver's compartment. The driver's compartment (cab) shall have manufacturer's current standard doors on each side. Doors shall be equipped with crank-regulated, safety glass windows. The doors shall be furnished with locks, operable from inside the vehicle with mechanical linkage, with both doors equipped with external key-operated locks. An individual upholstered driver's seat and an auxiliary seat shall be furnished. The driver's seat shall be adjustable. Seat belts shall be furnished for both seats. Outboard seats shall have combination pelvic and upper torso restraint seat belts. A domelight and ventilator(s) shall be provided. A full headliner shall be provided in the driver's compartment.

3.5.2.2 Curbside door(s). A single sliding door or split type double doors shall be provided on the curbside. The door(s) shall be furnished with lock(s).

3.5.2.3 Rear doors. Split type double rear doors shall be provided in the rear. The doors shall be provided with safety glass windows in the upper section. The doors shall be furnished with locks.

3.5.2.4 Utility shelves. Utility shelves with small parts bins shall be provided in the interior of the van body. One shelf unit shall be provided on the streetside wall and one on the curbside wall, near the rear of the van body over the wheelhousings. Shelf units shall not obstruct operator entrance and exit through the curbside or rear doors. Shelf units shall be not less than 1220 mm (48 inches) in length, 1070 mm (42 inches) in height and 300 mm (12 inches) in depth. Shelf units shall be securely bolted to the body floor and to the van sidewall braces. Each shelf unit shall have not less than three tray type shelves of not less than 18 gage (1.214 mm) (0.0478 inch) steel. Shelves shall be vertically adjustable and shall be anchored in place to withstand road vibration and bounce. Each shelf surface shall be designed so that any area of one square foot shall be capable of sustaining a live load of 45 kg (100 pounds) without permanent deformation. Shelf unit end panels shall be not less than 16 gage (1.519 mm) (0.0598 inch) steel. Not less than 20 adjustable shelf and parts bin dividers shall be furnished.

* 3.5.2.5 Van lighting. In addition to the lighting specified in 3.4.2.5, the following lighting and wiring shall be furnished:

(a) A domelight in the rear of the van body to illuminate the utility shelf unit area. The light shall be operable from a switch mounted on the instrument panel and by a rear door operated switch;
(b) Two 115 mm (4-1/2 inch) minimum diameter sealed beam floodlights with universal mounts, one on each side on the van roof at the front, with controls on the outside and operable by an operator from inside the driver's compartment through an open window for illuminating the work area;

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(c) An amber rotating vehicular warning light in accordance with MS 51317-4. The rotating light shall be mounted with reinforcement at the point of attachment on the vehicle centerline for 360 degree visibility. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch;

(d) Demountable projecting load light on the end of the boom, with a service line and socket when the end of the boom extends beyond the rear of the vehicle. A receptacle for the light on the rear of the body shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18;

(e) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292. All wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body frame work.

3.5.2.6 Basket access step(s). A safety tread steel plate step(s) shall be furnished to afford the operator easy access to the basket. The step(s) shall be not less than 250 mm (10 inches) deep and shall be permanently attached to or become a part of the rear bumper specified in 3.4.14.

3.5.2.7 Turntable pedestal mounting. The turntable pedestal shall be mounted on the truck frame inside the van body just to the rear of the driver's compartment and shall extend through the roof. The roof opening shall be sealed with a neoprene gasket(s). The pedestal mounting shall provide not less than 510 mm (20 inches) of clearance from the side door.

* 3.5.3 Style 3 flatbed trucks. Style 3 flatbed trucks shall be equipped with a cab and a flatbed type body having the nominal dimensions of 2440 mm (8 feet) wide and 2440 mm (8 feet) long for class A and 2440 mm (8 feet) wide and 2740 mm (9 feet) long for class B and C. The turntable pedestal shall be securely mounted to the truck chassis frame either directly or through a suitable subframe, and shall be located in the loadspace. The subframe and floor shall not be welded to the truck chassis frame. The loadspace floor shall be of not less than 4.8 mm (0.1875 inch) safety tread steel deck plate with suitable steel subframing. Any area of one square foot shall be capable of sustaining a live load of 227 kg (500 pounds) without permanent deformation.

3.5.3.1 Cab and bulkhead. Style 3 flatbed trucks shall be equipped with a cab and with a bulkhead as specified in 3.5.1.1 and 3.5.1.2.

3.5.3.2 Stirrup steps. Stirrup steps shall be provided at the rear of the flatbed to allow for easy access to the loadspace area.

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3.5.3.3 Tool boxes. An underslung weatherproof tool box shall be provided on both sides of the flatbed, extending from the front of the platform to the front of each rear wheelwell. The length, height and depth of the tool boxes shall be the maximum practicable. Each tool box shall have provisions for locking with a padlock or key operated lock. Tool boxes shall be fabricated of not less than 14 gage (1.897 mm) (0.0747-inch) sheet steel.

* 3.5.3.4 Flatbed body lighting. In addition to the lighting specified in 3.4.2.5, the following lighting and wiring shall be furnished:

(a) Lights on universal mounts and of sufficient candlepower to illuminate body loadspace and all equipment contained therein and all controls;
(b) Two 115 mm (4.5 inch) minimum diameter sealed beam floodlights with universal mounts shall be installed, one on each side on the top of the cab, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area;
(c) An amber, rotating, vehicular warning light in accordance with MS 51317-4. The rotating light shall be mounted with reinforcement at the point of attachment on or near the left front corner of the cab for 360 degree visibility. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch;
(d) A demountable projecting load light on the end of the boom, with a service line and a socket when the end of the boom extends beyond the rear of the vehicle. A receptacle on the rear of the body for the light shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18;
(e) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292. All wiring exposed to possible physical damage, such as wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body framework.

3.5.3.5 Mounting. The body mounting shall conform to 3.5.1.6.

3.6 Aerial boom types and components. Type I aerial boom shall be of the two or three stage telescopic type. Type II aerial boom shall be of the articulating folding arm type with a limited angle travel of not less than 0 to 125 degrees between booms. Type III aerial boom shall be of the articulating folding arm type with a limited angle travel of 90 to not less than 155 degrees travel between booms. Type IV aerial boom shall be of the articulating folding arm type, with an overcenter angle travel of not less than 0 to 180 degrees between booms. Boom travel of the lower arm on types II, III and IV shall be a minimum of 75 degrees vertical to the maintenance

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body platform or to the van roof. Aerial boom structural components shall have a calculated design stress, not greater than that allowed by ANSI A92.2 Outriggers or torsion bar stabilizers shall be furnished when required to meet the stability requirements specified herein and in ANSI A92.2.

3.6.1 Turntable. The turntable assembly shall be rotated by a hydraulically operated motor. The turntable shall be constructed to eliminate all side thrusts on the motor shaft.

3.6.2 Fiberglass reinforced plastic. The fiberglass boom(s), basket and components shall consist of extruded, molded or spiral-wound fiberglass. Each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be from a combination of multiple layers of sterile electrical grade fiberglass roving cloth, mat or strands with high impact strength resins with weathering and fire retardant characteristics. The insulated boom(s) shall have a hard, smooth surface inside and outside to prevent contamination and to preserve dielectric characteristics. When gelcoat is applied, the thickness of the gelcoat shall be not less than 0.16 mm (0.006 inch) and not more than 3.8 mm (0.030 inch). The parts shall be free from uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity and other similar defects. The entire surface shall be smooth, fully cured and free of tackiness. Percentage of glass by weight shall be approximately uniform. The plastic construction insulation shall meet or exceed the requirements of 3.6.3.

* 3.6.3 Dielectric properties. The complete aerial platform (aerial device) shall conform to and shall pass the design, quality assurance and qualification tests required for a category A or category B aerial device rated at 69 kV in accordance with ANSI A92.2, without exceeding the maximum attainable boom currents specified in ANSI A92.2 for a 69 kV rating. Any mechanism within or along the insulated boom(s) shall be made of insulating materials, or so insulated as to demonstrate the same dielectric properties as the boom(s).

3.6.3.1 Basket. The single wall fiberglass basket shall withstand a minimum of 30,000 volts ac, 60 Hz, without evidence of dielectric puncture (see 6.3). A decal stating "NO INSULATION VALUE WITHOUT LINER" shall be applied on each basket. The liner shall withstand a minimum of 50,000 volts ac, 60 Hz, without evidence of dielectric puncture. The double shell, insulated, fiberglass basket (see 3.6.5.1.1) shall withstand a minimum of 80,000 volts ac, 60 Hz, without evidence of dielectric puncture.

* 3.6.3.2 Dielectric test electrode(s). Except for type I telescopic boom, test electrode(s) shall be installed on the insulated boom section(s) in accordance with ANSI A92.2 Test electrode(s) shall consist of two collector bands, to match the inner and outer contours of the insulated boom section(s). The test electrode(s) and collector bands shall be arranged to

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enable monitoring dielectric leakage of the upper fiberglass boom, hydraulic hoses, hydraulic fluid and leveling devices. Collector bands shall be of a conductive material secured to the insulated boom section(s). The electrode(s) shall incorporate a terminal for test circuit connection. A dual-range meter of 2.0 milliampere and 200 microamperes shall be furnished in a leather or heavy duty vinyl carrying case and stowed in a protected location for use by the operating personnel. An instruction plate shall be provided and shall include the maximum dielectric leakage permitted and shall describe the dielectric test procedures. The basket liner(s) and booms shall be tested in accordance with ANSI A92.2.

* 3.6.4 Boom assembly. Type I boom assembly shall be of the hydraulic telescopic type. The boom assembly for types II, III and IV shall be of the hydraulically articulated folding arm type. The boom assembly shall be attached through the rotating turntable to a pedestal using self-alining bearings and shaft or shall be machined to assure alignment of the shaft and bearing surface. Under dynamic conditions, the boom assembly shall support the specified load placed in the aerial basket in any extended position without permanent deformation of any component in or along the boom(s) or its supporting structure.

3.6.4.1 Boom construction. The boom(s) shall be designed, engineered, and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to, or inside, the boom(s) shall be positioned for easy inspection, ready maintenance or possible replacement and shall have maximum protection against damage afforded by the configuration of the boom(s). The boom(s) shall be attached to the aerial basket and to the turntable by replaceable shafts. Rotating bearing surfaces of the boom at the turntable, and at the location of the lower and upper booms for types II, III and IV which are subjected to continuous wear, shall be provided with replaceable bearings or bushings. Lubrication fittings or permanently lubricated components shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent with the design. When completely closed boom(s) is furnished, means shall be provided within the boom(s) to absorb moisture. Boom(s) shall be provided with inspection openings at critical areas. Boom(s) wrapped for insulation purposes shall not be acceptable.

* 3.6.4.2 Boom operation. The rotating turntable and boom assembly shall be hydraulically actuated by a separate mechanisms and shall operate independently or in any combination of raising, lowering, and rotating at the same time. When in combination, hand metering of the controls may be necessary. The boom operating mechanism shall be capable of limiting the lifting capabilities from rest position within safe limits of design stress to prevent overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests. Upon completion of all testing, the hydraulic system pressure shall be adjusted to the

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manufacturer's recommended setting for the model vehicle furnished. The vehicle shall be inherently designed or provided with a device to prevent the boom(s) reaching a position that would cause the vehicle to become unstable.

3.6.4.3 Boom stowage. There shall be provisions to stow the boom(s) securely for normal storage, travel and shipment of the vehicle. Type I boom shall retract to an essentially horizontal position. Type II and type IV booms shall fold to an essentially horizontal position. Type III booms may be stowed with the lower boom in an essentially horizontal position and the upper boom in an essentially vertical position extending downward. Saddles or a bumper mounted boom rest shall be provided to support the boom and basket assembly in the stowed position. Tiedown or hold-down device(s) shall be provided to prevent unactuated movement of the boom(s) when in transit. Support and securement provisions shall prevent vertical and side movement of the boom and basket assembly in transit. Boom saddles furnished shall be sufficiently padded to prevent damage to the boom(s). The supports shall not interfere with the entrance and exit of personnel to the maintenance body loadspace or the van interior.

3.6.5 Aerial basket. A one-man fiberglass basket shall be furnished and shall have a load capacity of not less than 136 kg (300 pounds). Minimum interior dimensions shall conform to ANSI A92.2, Identification S-1 or S-4. The basket shall be fully enclosed on five sides. The basket shall have a molded-in or bonded on step, protruding toward the outside, located to permit easy entrance and exit of the operator. The fiberglass basket shall be provided with a guard to protect the controls.

* 3.6.5.1 Basket with liner. A nonconductive basket liner shall be provided. Dielectric strength performance requirements of the basket and liner shall be as specified in 3.6.3.1. The liner shall enclose the basket interior. The liner shall be easily removable from the basket for cleaning and for testing the dielectric capacity. The floor of the liner shall be provided with a removable, nonconductive scuff mat.

3.6.5.2 Flooring. The basket shall have a reinforced floor. No point in the flooring shall deflect more than 6.4 mm (0.25 inch) nor be permanently deformed when subjected to a vertical load of 136 kg (300 pounds) covering an area of 160 sq cm (25 square inches).

3.6.5.3 Basket cover. A waterproof basket cover and means to retain the cover on the basket, while in stowage or stowage-in-transit position, shall be provided on the basket and also may enclose the controls; otherwise, a cover shall be provided to completely enclose all control mechanisms. Provisions shall be made to store the cover(s) in the body.

3.6.5.4 Safety belt. A lineman type safety belt with positive-lock quick release features shall be provided. The belt shall have a design strength of 910 kg (2,000 pounds). The safety belt shall be designed and

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positioned to allow freedom of movement of the workman in the basket. Means shall be provided to attach fasteners to the end of the boom or basket bracket and to prevent fastener attachment being pulled through the plastic wall, in the event of leveling system failure. Fasteners and attachments shall be corrosion-resistant.

3.6.5.5 Tool tray. A fiberglass or polyethylene tool tray shall be provided to fit over the edge of the basket. Provisions shall be made to secure and stow the tray in the body.

3.6.5.6 Basket leveling system. An automatic basket leveling system shall be provided for keeping the basket horizontal in any boom position with rated load positioned on the basket floor. The leveling system shall be designed with minimum stretch due to loading the basket. Provisions for adjusting the level of the basket shall be provided. The leveling system components shall be provided with maximum protection based on design features and safety. Provisions to prevent tipping of the basket in case of failure shall be provided. The system shall be insulated as specified (see 3.6.3). The leveling system shall be automatically operated in all extended positions that the boom(s) is capable of attaining. The basket shall be capable of being released and easily tilted over by one man, near ground level, manually or hydraulically, to remove a disabled operator or to dump out water.

3.6.6 Controls. Two complete sets of hand-operated controls for maneuvering the basket shall be provided. One set of controls shall be provided at the basket and shall be located at the side of the basket in a position easily accessible to and not behind the operator facing the work area. The lower set of controls shall be located at the base of the boom on the maintenance and flatbed trucks or inside the rear door of the van (see 3.6.6.1 and 3.6.6.2). Lower controls shall be capable of overriding the controls at the basket. Smooth operation, with no lurching of the boom assembly, shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. All boom controls shall be the dead-man type operable with one hand but whereby two separate actions must be taken to activate the control. A single lever control may be furnished in the basket. The placement of the controls shall prevent inadvertent actuation by the operator. The controls shall be identified as to their function. Identification markings shall be on a photoetched, corrosion-resistant plate. Shockproof gages and instruments shall be provided and shall be weatherproof.

3.6.6.1 Maintenance and flatbed truck lower controls. Maintenance and flatbed truck lower controls shall be located at the base of the boom and shall be easily accessible from the floor of the body, or a platform shall be provided for the operator at the base of the boom. Lower controls shall be placed to minimize the possibility of use by unauthorized personnel. Lower controls shall be contained in an enclosed box(es) with side-hinged cover(s), furnished with a hasp for locking with a padlock.

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3.6.6.2 Van lower controls. The lower control box on the van shall be mounted inside the rear curbside door on a slide or clip mount. The control box shall be removable for use outside the van and shall be furnished with a cable not less than 3050 mm (10 feet) in length. Clip, hook or stowage space shall be furnished to secure the cable when the control box is stowed on the rear door mount. The control box shall be key-operated. Interior wiring shall be routed through the van to the rear door in a safe manner to prevent interference with personnel or damage to the wiring.

3.6.7 Hydraulic system. A hydraulic fluid, nonconductive system consisting of all components required to perform all hydraulic operations of the boom assembly with the degree of control specified herein shall be provided. Mechanisms and reservoirs shall be positioned or shielded to prevent contact of hazardous fluid with heated surfaces or electrical apparatus. An accumulator may be used as a component of the hydraulic system to act as a pressure storage reservoir, maintaining the pressure under continuous service.

3.6.7.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and to prevent hydraulic fluid leaks from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the boom shall be provided with pilot check or holding valves to prevent descent of the boom assembly in case of hydraulic pressure loss. Means shall be provided to install pressure gages for testing hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome-plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.6.7.2 Hydraulic lines and fittings. Hydraulic lines shall be located to provide maximum ease of removal and installation, visual inspection for failures and maximum protection from damage during operation of the boom assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptors to flared type fittings, or tapered pipe thread with dry seal fittings shall be furnished. Hydraulic lines and fittings shall be air- and liquid-tight and shall be installed to prevent vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines. A manual type bleeder shall also be provided as on-vehicle-equipment on closed-center, non-self bleeding systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable.

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* 3.6.7.3 Hydraulic pump. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The pumping system shall include an adjustable pressure relief valve, unloading valve or pressure compensator with a range of adjustment which allows raising or lowering, extending or retracting of the boom and basket with any working or test load specified herein. A pressure gage shall be provided on the discharge side of the hydraulic pump.

* 3.6.7.4 Pump drive systems. For style 1 maintenance and style 3 flatbed trucks with manual transmissions, the hydraulic pump shall be driven by a power takeoff (PTO) unit attached to the truck chassis manual transmission. For all other styles and transmission types, the hydraulic pump shall be driven by the engine crankshaft or the engine fan belt. For emergency use, an auxiliary hydraulic pump, driven by an electric motor powered from the vehicle chassis electrical system, shall be provided to rotate the turntable, retract the outriggers (if furnished), lower the boom and place the booms in stowed position in the event of primary power source failure. The system shall be equipped with cut-in devices and controls at the basket and at the lower control station. The system shall provide adequate pressure and flow capacities, for operation when the primary power source fails. When specified (see 6.2), in addition to the PTO, crankshaft or fan belt drive, the pump shall be driven by a rechargeable electrical system (see 3.6.9). When specified (see 6.2), in addition to the PTO, crankshaft or fan belt drive, the pump shall be driven by an auxiliary engine (see 3.6.10). In all cases, a means of rapid conversion from one power source to the other, from within the basket and from the ground, shall be provided. Chain drives shall not be used in any system.

3.6.7.5 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided, with sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be adequately baffled. The reservoir shall be provided with a filler opening, a removable fill opening oil strainer and a vented filler cap, located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of the tank during movement of the vehicle. Filler cap shall be attached to the tank by a corrosion-resistant chain or cable. Provision shall be made for inspection and cleaning of the interior and for draining of the reservoir. The reservoir drain shall be located so that fluid will not drain on other components of the vehicle and shall allow recovery of the fluid in a container. The reservoir shall have a corrosion-resistant caution plate to indicate the type of hydraulic fluid furnished and the maximum fluid level in the reservoir with boom retracted for travel position and the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to observe the level of fluid. The tank shall be clean.

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3.6.7.6 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

3.6.7.7 Oil filter and strainers. A full flow type, 10-micron discharge oil filter shall be provided in the return line to the reservoir or in the pressure line immediately after the pump. A 100-mesh suction line strainer shall be provided on the suction side of the hydraulic system. The filter and the suction strainer shall be easily accessible for periodic cleaning or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow maximum rated pump discharge.

3.6.7.8 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage or approach to the flash point of the fluid during all operating conditions of the vehicle and use of the aerial basket for long periods of time. Excessive heating of the fluid (more than 13°C (55°F) above ambient in an ambient of 52°C (125°F) or equivalent at lower temperature) shall not be acceptable under any operating conditions. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in the fluid temperature.

* 3.6.8 Basket engine starter and cutoff. When a power takeoff driven system is required, an engine starter and cutoff remotely operated from the basket shall be furnished. The engine starter and cutoff shall not reduce the insulating characteristics specified in 3.6.3.

* 3.6.9 Rechargeable electrical system. When a rechargeable electrical system is required (see 3.6.7.4), the electric motor shall be mounted in conjunction with the hydraulic pump. The motor shall have adequate output for full compliance with all performance and lifting and speeds specified herein. Mounting shall minimize transfer of vibration. The electrical system shall include:

(a) A minimum 12-volt dc motor
(b) Industrial battery(s) of not less than 450 ampere-hour capacity
(c) Provisions for charging battery(s) from the vehicle alternator (see 3.4.2.3)
* (d) A 110- or 220-volt, single phase battery charger with a dc output capable of recharging the battery(s) in (b) above in not more than 12 hours
(e) Hour meter
(f) Battery charge indicator
(g) Pressure switches and control circuits.
* (h) Battery isolator connected to the chassis charging system which prevents a drain of the chassis battery during boom operations.

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3.6.10 Auxiliary engine. When an auxiliary engine is required (see 3.6.7.4), the engine shall be mounted on the truck body platform in conjunction with the hydraulic pump. The engine mounting shall minimize transfer of vibration. The auxiliary engine shall be air-cooled, and shall operate on the same fuel as the vehicle chassis engine. The auxiliary engine shall have sufficient output power to drive all operating components other than the vehicle. Gasoline engines shall be capable of warranted operation on unleaded fuel.

3.6.10.1 Engine components. The engine shall be a four-cycle gasoline engine or a two- or four-cycle diesel engine (same type as chassis engine) equipped with at least the following components:

(a) Distributor or magneto type ignition (gasoline engine only)
(b) Starter, 12-volt
(c) Resistor type spark plugs (gasoline engine only)
(d) Air cleaner
(e) Automatic or manual choke (gasoline engine only)
(f) Engine hour meter
(g) When specified (see 6.2), a power plant heater having a 300-watt immersion type heater with 77°C (170°F) to 90°C (195°F) thermostat shall be installed in the oil pan through any convenient opening.

3.6.10.2 Engine control panel. The engine throttle, a key-operated ignition (starting) switch, and a palm button starter shall be installed in a weather and waterproof cabinet on a control panel, and each shall be identified as to its function. The cabinet shall be provided with a side-hinged cover with a spring type latch and provisions for locking with a 1-inch padlock.

3.6.10.3 Electrical system. The auxiliary engine electrical system shall utilize the truck-chassis battery and shall be in accordance with Federal Motor Carrier Safety Regulations 393.30 through 393.33.

3.6.10.4 Fuel pump. A fuel pump shall be furnished on the engine. When lifting fuel against a head of 1220 mm (4 feet), the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided, either as a component of the fuel pump or between the fuel pump and the carburetor.

3.6.10.5 Governor. A variable speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually-operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

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3.6.10.6 Auxiliary engine fuel supply. Unless an auxiliary engine fuel tank is specified (see 3.6.10.6.1), the auxiliary engine shall utilize the vehicle's main engine fuel supply (see 3.4.3). A shut-off valve shall be provided in the line to the power source and shall be located in an accessible location. Fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.6.10.6.1 Auxiliary engine fuel tank. When specified (see 6.2), an auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a corrosion-resistant chain or cable. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. Drainage shall be provided to prevent spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the line to the engine and shall be adjacent to the fuel tank. A corrosion-resistant plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap.

3.6.10.7 Exhaust system. Means shall be provided to vent exhaust fumes away from the cab and body. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the auxiliary engine exhaust system. Model 4x4 auxiliary engine exhaust systems shall be furnished with a spark arresting muffler (see 3.4.4).

3.6.11 Hydraulic tools. For style 1 maintenance and style 3 flat-bed trucks, the following hydraulic tool outlet and, when specified, hydraulic tools shall be provided.

3.6.11.1 Hydraulic tool outlet. A hydraulic tool outlet shall be provided in the curbside cabinet or in a protected position on the rear, under the platform or on the curbside of the turntable pedestal. Discharge port(s) and return port(s) shall be provided with a corrosion-resistant cap, secured with a corrosion resistant chain or cable, to protect the threads and the valve from contamination. The tool outlets shall use hydraulic fluid pressure from the hydraulic system of the aerial platform. One tool circuit shall be furnished, with quick disconnect, dripless couplers for attaching the pressure and return hose lines for the motor driven hydraulic tools. The oil flow for the tools shall be regulated by control valve(s) to meet the flow requirements of the hydraulic tools. The high system pressure (working pressure) shall be not less than 11.0 megapascals (MPa) (1600 psi) and the back-pressure in the return line shall not exceed 1.05 MPa (150 psi).

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3.6.11.1.1 Ground tool hose. Except when a hydraulic hose reel is specified (see 3.6.11.11), two 15 200 mm (50-foot) lengths of hydraulic wire braid hoses with quick disconnect, dripless couplers on both ends for use on ground level shall be furnished. Warning plate markings shall warn that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hose shall be furnished with dust caps. Hose fittings shall be compatible with the tool outlet(s) specified in 3.6.11.1 and the tools specified in 3.6.11.3 through 3.6.11.8.

3.6.11.2 Basket hydraulic tool outlet and hose. A hydraulic tool outlet shall be furnished in the basket. An 2440 mm (8-foot) dual hydraulic hose, for use in the basket, shall be furnished. The hose shall be a flexible, nonconductive type with quick disconnect dripless fittings on each end. Hose fittings shall be compatible with the basket outlet and the tools specified in 3.6.11.3 through 3.6.11.8. The hose ends shall be provided with dust caps.

3.6.11.3 Hydraulic capstan hoist. When specified (see 6.2), a hydraulic capstan hoist shall be furnished. The drum shall be one direction drive and shall have a backstop on the output shaft to prevent reverse travel. Lowering will be accomplished by letting the rope slip on the drum. The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than on a round pole. The hoist shall have a rated single line pull capacity of not less than 567 kg (1,250 pounds) on the bare drum. The hoist shall have a remote control foot-operated open-center valve.

3.6.11.4 Hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing or removing lag screws and other hardware. The wrench shall have a 16 mm (0.625 inch) standard bolt size capacity with a 13 mm (0.500 inch square) drive. The tool shall be capable of developing not less than 270 Joules (J) (200 foot-pounds) of torque with at least 1,350 impacts per minute. A quick change drill chuck shall be furnished.

3.6.11.5 Hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall have a chipper type chain and shall be suitable for cutting trees up to 380 mm (15 inches) in diameter.

3.6.11.6 Hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be lightweight (approximately 5 kg (10 pounds)) and shall have a cutting capacity of not less than 230 mm (9 inches). The pole shall be approximately 1520 mm (5 feet) long and shall have a fiberglass type of body to provide good insulation.

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3.6.11.7 Hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall have a capacity of not less than 81 mm (3.250 inches) and shall operate at approximately 2700 meters per minute (8,900 feet per minute). The pole shall be approximately 1830 to 2130 mm (6 to 7 feet) long and shall have a fiberglass type of body to provide good insulation. The saw shall have at least a 44-tooth blade.

* 3.6.11.8 Hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall have a capacity of not less than 50 mm (2 inches) in diameter. It shall be mounted on a fiberglass type pole to provide good insulation and shall be approximately 2130 mm (7 feet) long.

3.6.11.9 Hydraulic fittings. Quick disconnect dripless fittings shall be provided on each hydraulic tool. The fittings shall be compatible with the hoses specified in 3.6.11.1.1, 3.6.11.2 and 3.6.11.11.

3.6.11.10 Hydraulic tool stowage. Hydraulic tools, when furnished, shall be stowed on brackets inside a curbside cabinet. The brackets shall be provided with retaining devices to secure the tools.

3.6.11.11 Hydraulic hose reel. When specified (see 6.2), a heavy duty retractable hydraulic hose reel and hose shall be furnished. The hose reel shall be furnished with a mounting plate secured in the vehicle cabinet or in an accessible location on the vehicle platform. The hose reel shall be furnished with a weatherproof cover for protection when not in use. The hose reel shall be capable of withstanding not less than 12.4 MPa (1,800 psi) line pressure. The hose is intended for use in the aerial basket and thus the hose and hose reel shall be capable of meeting the dielectric properties specified in 3.6.3. The hose reel shall be capable of retraction. The hose reel shall be marked stating that the hose shall be walked back to the reel to prevent damage. Hose fittings shall be of the quick-disconnect dripless type and shall be compatible with the tool outlets specified in 3.6.11.1 and 3.6.11.2 and with the tools specified in 3.6.11.3 through 3.6.11.8. The following accessories shall be furnished:

(a) Flow control valve(s)
(b) Line check valve(s)
(c) Hose ball stop
(d) Four-way roller hose guide
(e) Dual hose, 15 200 mm (50 feet) in length, with dust caps.

3.7 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

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3.8 Stress analysis. A stress analysis for each type, class, style, and size of vehicle under contract will be required by the contract. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall show the location of the center of gravity of each component relative to the vehicle axles (travel and critical working positions). In addition, the data shall include the value of stresses at the base of the boom, at the hinge area between the extension boom section and the basket for type I, at the hinge area between the upper and lower boom sections for types II, III, and IV, at the hinge area between the upper boom section and the basket for types II, III, and IV, and at the center of gravity of each section of the boom. The analysis shall prove conformance to ANSI A92.2.

3.9 Dissimilar metals. All dissimilar metals throughout the vehicle shall be insulated from one another to prevent galvanic action or electrolytic action.

3.10 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. Vehicles shall be lubricated, prior to delivery, with an applicable grade of standard lubricant for the ambient air temperature encountered at the delivery point.

* 3.11 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

* 3.11.1 Welding. Adequate preparation of metal prior to the welding of the body and aerial tower components shall be provided. The surface of parts to be welded shall be free of rust, scale, paint, grease, and other foreign matter. Welding rods (electrodes) shall be compatible with the metal being welded. All critical arc welds shall be magnetic particle or dye penetrant inspected. The following defects shall not be acceptable:

(a) Slug or plug welding
(b) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments, and vertical down welding
(c) Unremoved slag and spatter
(d) Cracks
(e) Spot, tack, or intermittent weld for strength.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.12.2 through 3.12.3.

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3.11.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body and aerial tower components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approved specimens shall be retained at the contractor's plant.

3.11.3 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.11.4 Factors. The factors listed below which are applicable shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to

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ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle preliminary inspections.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to start of fabrication of the sub-floor, turntable and boom assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

* 4.3.2 In-process inspection of body and components. The first complete assembled body and components (subframe, turntable, boom assembly and basket) shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be conducted to evaluate workmanship and to determine conformance to contractual requirements.

* 4.3.2.1 Structural inspection. The vehicle shall be tested in accordance with 4.4.8 and 4.4.9 and dye penetrant or magnetic particle inspection shall be performed on the critical areas of the welds prior to painting. A water soluble penetrant inspection shall be performed on all critical plastic areas. The water soluble penetrant or magnetic particle inspections shall provide means to detect crack indications after dynamic and static loads.

4.4 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on the front and rear axles shall be

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computed by the contractor and verified by the Government to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contractual requirements. The distribution of weight shall be computed by using curb weight, the operator weight at 80 kg (175 pounds), and a payload considered to be evenly distributed throughout the length of the vehicle body, to provide the required GVW.

4.4.2 Road test. The first production vehicle shall be examined and road tested by the contractor to assure that the vehicle shall operate in accordance with contractual requirements. The vehicle, loaded with payload and all the equipment specified herein, shall be driven at speeds up to at least 80 km/h (50 mph), for a minimum distance of 16 km (10 miles). The distance traveled shall consist of a highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 16 km (10 miles) of travel at least one sudden stop shall be accomplished. Upon completion of the road test all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all doors shall be opened and shut; there shall be no evidence of mud or dust inside, and there shall be no evidence of racking of the body.

4.4.3 Truck body treatment and painting. For style 1 and style 3 vehicles, the certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to the Government representatives at the first production vehicle inspection.

* 4.4.4 Upsetting and range of movement test. With the truck or van resting on approximately level ground and not fastened to any artificial base, with one and one half times the specified payload (see 3.6.5) shall be placed on the basket floor or suspended from the basket. The basket shall be extended to its full reach and exercised through the full range of traverse and elevations indicated in 3.3.4 to see if any tire lifts off the ground.

* 4.4.4.1 Stability on slopes test. The tests specified in 4.4.4 shall be repeated, except the truck shall be resting on a 5 degree slope in positions specified in 3.3.4.1 and with the basket loaded with 1-1/3 times the rated payload (see 3.6.5).

4.4.5 Wind test. With the rated capacity placed in the basket (see 3.6.5), the boom shall be extended to maximum height and shall be rotated to that position whereby the greatest area of the basket is parallel to the longitudinal centerline of the vehicle. The specified wind load, generated by artificial means or by an equivalent simulated load, shall be applied to the vehicle from either side (see 3.3.5). The contractor shall make available calculations or a certified test report.

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4.4.6 Speed test. With the vehicle resting on approximately level ground, the basket shall be loaded with specified load (see 3.6.5). The basket shall be traversed and elevated to determine conformance to the requirements in 3.3.6, 3.3.7 and 3.3.8.

* 4.4.7 Creep test. With the vehicle longitudinal on a 5 percent grade, with specified load (see 3.6.5) in the basket, and with the hydraulic fluid stabilized at or near ambient temperature, the basket shall be extended to the side with the boom at 45 degrees for type I and with the lower arm in vertical or nearest capable vertical position and the upper arm in horizontal position for types II, III, and IV. The position shall be held for at least 1 hour for maximum horizontal creep of 25 mm/h (1 inch per hour) and vertical settling of 50 mm/h (2 inches per hour), to determine conformance to the requirements in 3.3.9.

* 4.4.8 Dynamic conditions. With two times the specified payload (see 3.6.5) on the basket floor or suspended from the basket, the boom(s) shall be extended through various positions, as deemed necessary by the Government inspector, and with the turntable shall be rotated 360 degrees. Then, dye penetrant or magnetic particle inspection shall be performed on critical welds. A water soluble penetrant inspection also shall be performed on all critical plastic areas. The water soluble penetrant or magnetic particle inspections shall provide means to detect indications of cracks. A visual inspection shall be performed of all critical fiberglass components. Any evidence of cracks or deformation or of damaged fiberglass material shall be cause for rejection of the vehicle.

* 4.4.9 Static conditions. Type I boom shall be placed in its most vertical position with two times the specified load (see 3.6.5) placed in or suspended from the basket. Types II, III and IV shall be placed with the lower boom in its most vertical position, the upper boom horizontal and with two and two-thirds times the specified load (see 3.6.5) placed in or suspended from the basket. Then at least the following areas shall be inspected by dye penetrant or magnetic particle:

(a) Turntable, base, trunnion, and yoke at attachment of lower boom and struts
(b) Boom at cylinder clevis brackets (lower boom only on types II, III and IV)
(c) Lower boom attachment to upper boom (types II, III and IV only)
(d) Structural steel of boom(s)
(e) Yoke at attachment to basket and side brackets.

Any evidence of cracks or deformation shall be cause for rejection of the vehicle.

* 4.4.10 Dielectric tests. The vehicle shall be tested as specified in the Qualification Test of ANSI A92.2, for Category A and B aerial devices for a 69,000 volts (69 kV) unit rating.

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4.4.11 Performance test. The platform shall be subjected but not limited to the following tests:

(a) Operate equipment using each set of controls separately
(b) Check to ensure that the lower controls override the basket controls
(c) Check to ensure that the entire system can be operated smoothly and without jerk motions in any position
(d) Check to determine whether the basket remains level in all positions with the specified load in the basket
(e) Check to determine whether the basket is safely lowered from maximum reach position with rated payload on the basket floor to within 36 inches above the ground for type I and within 48 inches above the ground for types II, III, and IV with the hydraulic pump and engine not operating
(f) Check the remote control starter and cutoff on basket when furnished, to ensure starting and stopping engine with a specified payload on the basket
* (g) Check overload capabilities by attempting to lift from rest position a load that will result in exceeding the manufacturer's recommended hydraulic system pressure setting
(h) Check the hydraulic system including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

* 4.4.12 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 4.4.13 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract.

4.4.14 Failure. Failure of the first production vehicle to meet the requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrected.

* 4.5 Availability of ANSI A92.2. The contractor shall make available to the Government at the point of final acceptance and test, current copies of ANSI A92.2. The document shall be surrendered to the Government for the personal use of the Government representatives in conjunction with the contract for the duration of the contract.

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4.6 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.22 have been accomplished.

* 5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government, for distribution lines not exceeding 69,000 volts with the basket liner installed for utility line maintenance, and for replacement of lamps, tree trimming, painting, window washing, and many other types of overhead construction and maintenance work.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, style, size, and model required (see 1.2 and 1.2.4).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
* (d) Identification of appropriate service for marking (see 3.1.1.2).
(e) Concealed marking, if required (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.6).
(g) Tropical rustproofing, if required (see 3.1.1.6).
(h) Silicone brake fluid, if required and available (see 3.1.1.8).
(i) A trailer towing package, if required (see 3.1.1.9).
* (j) Drive hubs, for model 4x4 vehicle, if required (see 3.1.1.10).
(k) Continuous rotation, if required (see 3.3.8).
(l) Diesel engine, if required (see 3.4.1.2).
* (m) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.2).
(n) Power plant heaters and fuel warmers, if required (see 3.4.1.7).
(o) Alternator capacity, if other than as specified (see 3.4.2.3).
(p) Automatic transmission, if required (see 3.4.5.3).
(q) Traction control, if required (see 3.4.9.1).
* (r) Bias ply, low profile or other type tires, if required (see 3.4.10.1).

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(s) If a spare tire carrier is not required (see 3.4.10.2).
(t) A spare wheel or rim, if required (see 3.4.10.3).
(u) A spare tire assembly, if required (see 3.4.10.4).
* (v) Intermittent windshield wipers, if required (see 3.4.13).
(w) Tools, if required (see 3.4.15).
* (x) AM/FM radio, if required (see 3.4.21).
(y) Individual driver's and passenger's seats, if required for styles 1 and 3 (see 3.5.1.1).
(z) Styles 1 and 3 manual transmission vehicles, hydraulic system to be driven by rechargeable electrical system or an auxiliary engine (in addition to standard power takeoff drive) if required (see 3.6.7.4).
(aa) Styles 1 and 3 automatic transmission vehicles, hydraulic system to be driven by an auxiliary engine in lieu of a rechargeable electrical system, if required (see 3.6.7.4).
(ab) Type MIL-H-83282 hydraulic fluid, if required (see 3.6.7.6).
* (ad) An auxiliary engine power plant heater, if required (see 3.6.10.1).
(ac) An auxiliary engine fuel tank, if required (see 3.6.10.6.1).
(ae) Hydraulic capstan hoist, if required for styles 1 and 3 (see 3.6.11.3).
(af) Hydraulic impact wrench and drill, if required for styles 1 and 3 (see 3.6.11.4).
(ag) Hydraulic chain saw, if required for styles 1 and 3 (see 3.6.11.5).
(ah) Hydraulic chain saw, pole mounted, if required for styles 1 and 3 (see 3.6.11.6).
(ai) Hydraulic circular saw, pole mounted, if required for styles 1 and 3 (see 3.6.11.7).
(aj) Hydraulic pruner, pole mounted, if required for styles 1 and 3 (see 3.6.11.8).
(ak) Hydraulic hose reel, if required for styles 1 and 3 (see 3.6.11.11).

6.3 Basket insulation. The 30,000-volt basket insulation is required (see 3.6.3.1) to obtain a heavy duty basket devoid of thin areas and imperfections in the basket walls.

* 6.4 Subject term (key word) listing.

Chassis, truck
Truck, commercial
Nontactical truck
Turntable mounting
Aerial boom types
Boom construction
Basket leveling system
Auxiliary engine

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-62355B; TRUCK, MAINTENANCE: TRUCK, VAN: AND TRUCK, FLATBED: WITH TELESCOPIC OR ARTICULATING BOOM ONE-MAN AERIAL BASKET, 7600 TO 9140 MILLIMETER (25 TO 30 FOOT), MODIFIED COMMERCIAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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FOR NEW DESIGN

MIL-T-62355B
NOTICE 1
30 October 1995

MILITARY SPECIFICATION

TRUCK, MAINTENANCE: TRUCK, VAN: AND TRUCK, FLATBED: WITH TELESCOPIC OR ARTICULATING BOOM ONE-MAN AERIAL BASKET, 7600 TO 9140 MILLIMETER (25 TO 30 FOOT), MODIFIED COMMERCIAL

This notice should be filed in front of MIL-T-62355B, dated 6 November 1991.

MIL-T-62355B is inactive for new design and is no longer used by the Army except for reprocurement purposes.

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Army - AT

(Project 2320-0664)

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